

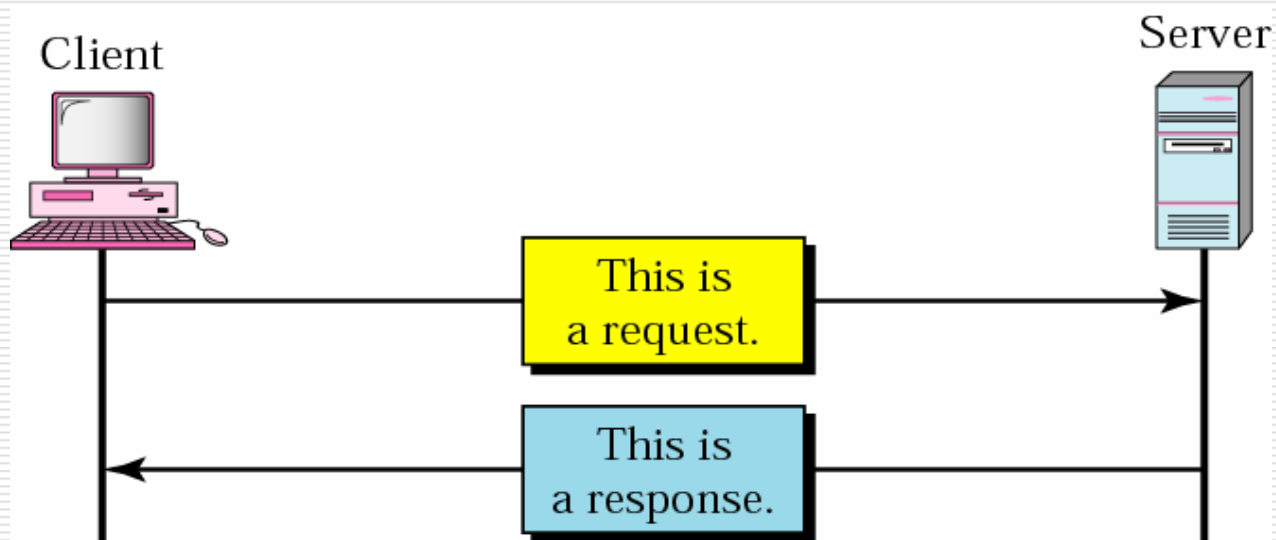
HTTP



Hyper Text Transfer Protocol (HTTP)

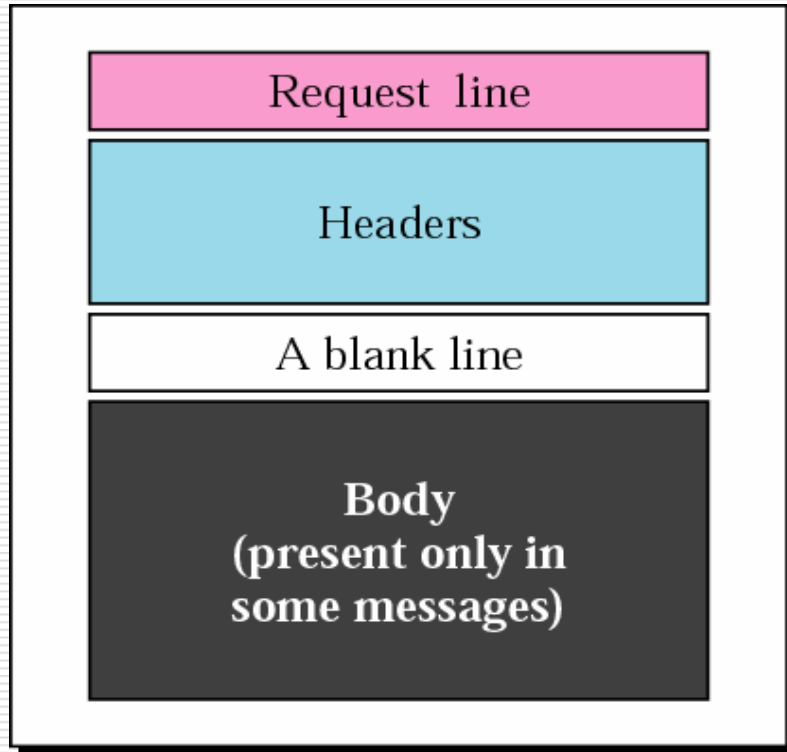
- Web's application layer protocol
 - client/server model
 - client: browser that requests, receives, “displays” Web objects
 - server: Web server sends objects in response to requests
 - HTTP uses the services of TCP on well-known port 80
-

HTTP transaction

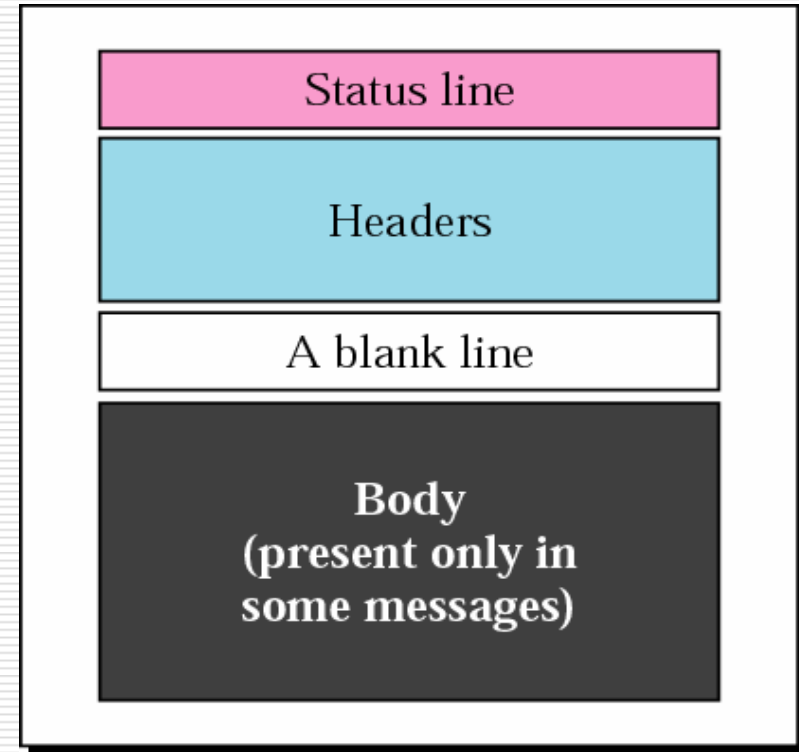


http is "stateless" server maintains no information about past client requests

Request message & Response message

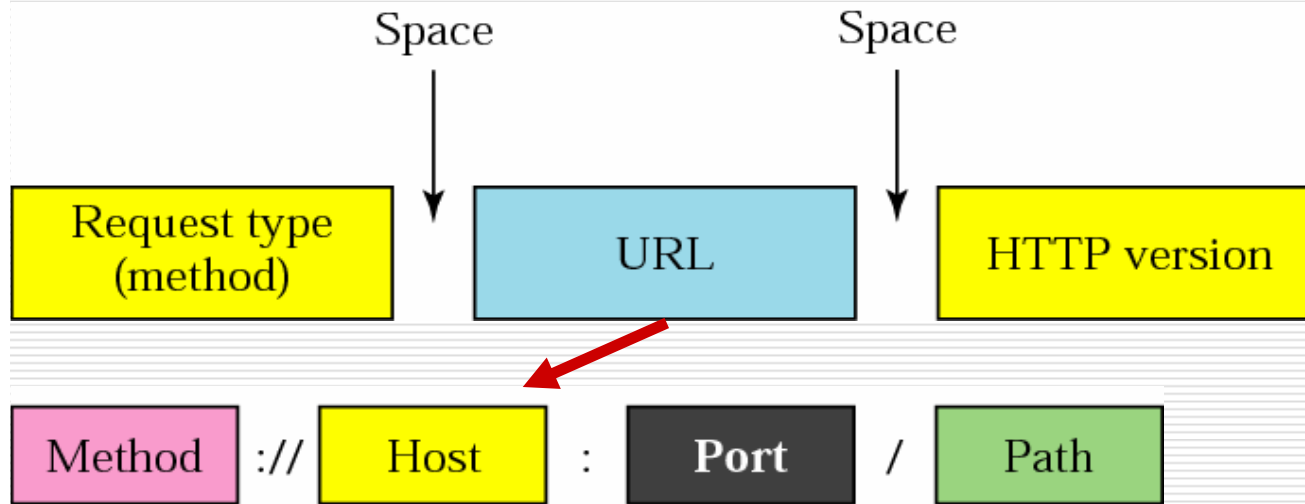


Request message

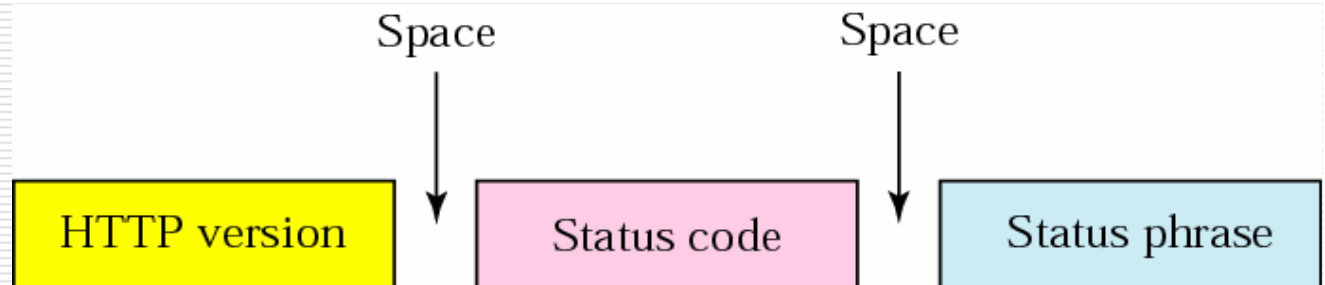


Response message

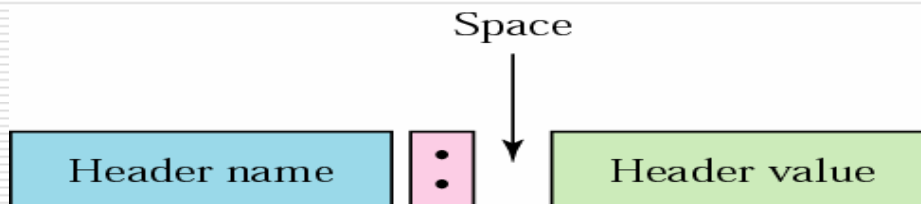
Request line



Status line



Header format



http message format: request

request line
(GET, POST,
HEAD commands)

header
lines

```
GET /somedir/page.html HTTP/1.0
User-agent: Mozilla/4.0
Accept: text/html, image/gif, image/jpeg
Accept-language: fr
```

Carriage return,
line feed
indicates end
of message

(extra carriage return, line feed)

http message format: *response*

The diagram illustrates the structure of an HTTP response message. It consists of three main parts: a status line, header lines, and the message body. Red arrows point from descriptive labels to their corresponding parts in the message format.

status line
(protocol
status code
status phrase) → **HTTP/1.0 200 OK**

header
lines → **Date: Thu, 06 Aug 1998 12:00:15 GMT**
Server: Apache/1.3.0 (Unix)
Last-Modified: Mon, 22 Jun 1998
Content-Length: 6821
Content-Type: text/html

data, e.g.,
requested
html file → **data data data data data ...**

Non-persistent and persistent connections

Non-persistent

- ❑ HTTP/1.0
- ❑ server parses request, responds, and closes TCP connection
- ❑ 2 RTTs to fetch each object
- ❑ Each object transfer suffers from slow start

But most 1.0 browsers use parallel TCP connections.

Persistent

- ❑ default for HTTP/1.1
- ❑ on same TCP connection: server, parses request, responds, parses new request,...
- ❑ Client sends requests for all referenced objects as soon as it receives base HTML.
- ❑ Fewer RTTs and less slow start.

Web Caches (proxy server)

Goal: satisfy client request without involving origin server

- user sets browser: Web accesses via web cache
- client sends all http requests to web cache
 - if object at web cache, web cache immediately returns object in http response
 - else requests object from origin server, then returns http response to client

